

A peek into the MSRE, six decades later: a hyper-fidelity simulation of the classical molten salt reactor

Rok Krpan^{1,*}, Carlo Fiorina¹, Kevin Clarno², Chiara Genoni¹, Cole A. Gentry², Sun Myung Park², Jean Ragusa¹

¹Texas A&M University, College Station, TX, USA; ²The University of Texas at Austin, Austin, TX, USA

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ABSTRACT

The Molten Salt Reactor Experiment (MSRE) was a pioneering nuclear reactor built and operated at the Oak Ridge National Laboratory to demonstrate the key principles of molten salt reactor technology. This work presents the first hyper-fidelity, fully heterogeneous simulation of the MSRE, resolving all reactor vessel internals without significant geometric simplifications or modeling assumptions. The steady-state behavior of the reactor is simulated by coupling high-fidelity neutronics and thermal-hydraulic analyses. Spatial power distributions were obtained using the Serpent2 Monte Carlo code, while detailed velocity, temperature, and density fields were computed with the foamForNuclear multiphysics code through conjugate heat transfer simulations. The simulation results, validated against experimental data, demonstrate the model's ability to accurately reproduce the reactor's design parameters and operational behavior.

Keywords: Molten Salt Reactor Experiment, Hyper-Fidelity Multiphysics Simulation, foamForNuclear multiphysics code, Serpent2 Monte Carlo code

1. INTRODUCTION

The Molten Salt Reactor Experiment (MSRE) was a pioneering nuclear reactor built and operated at the Oak Ridge National Laboratory (ORNL) from 1965 to 1969 to demonstrate the key features of the MSRs [1, 2, 3]. The MSRE was a graphite moderated nuclear reactor, which used liquid LiF-BeF₂-ZrF₄-UF₄ as both the fuel and the coolant. The MSRE reactor vessel is shown in Fig. 1, and Fig. 2 shows the cross section of the MSRE reactor core. The main components are the salt distributor (volute), the downcomer (annulus), the lower plenum with the anti-swirl vanes, the moderator grid support plates, the graphite lattice blocks, the graphite moderator (core blocks), and the upper plenum. The moderator region was composed of the graphite blocks with a square cross section, with longitudinally placed channels at each side. When these blocks were placed together into the reactor core, 1140 fuel passages were formed.

In the present work, a hyper-fidelity heterogeneous model of the MSRE vessel and its internals has been developed, without significant geometric and physical simplifications. The steady-state behavior of the reactor is simulated by coupling high-fidelity neutronic and thermal-hydraulic analyses. Power distributions are obtained using the Serpent2 Monte Carlo code [4], while detailed thermal-hydraulic simulations are performed with the foamForNuclear framework, a multiphysics finite-volume framework derived by merging the capabilities of the GeN-Foam multi-physics platform [5] and the OFF-BEAT fuel behavior and non-linear thermo-mechanics tool [6], and designed for application across a range of fission and fusion reactor technologies.

*rok.krpan@tamu.edu

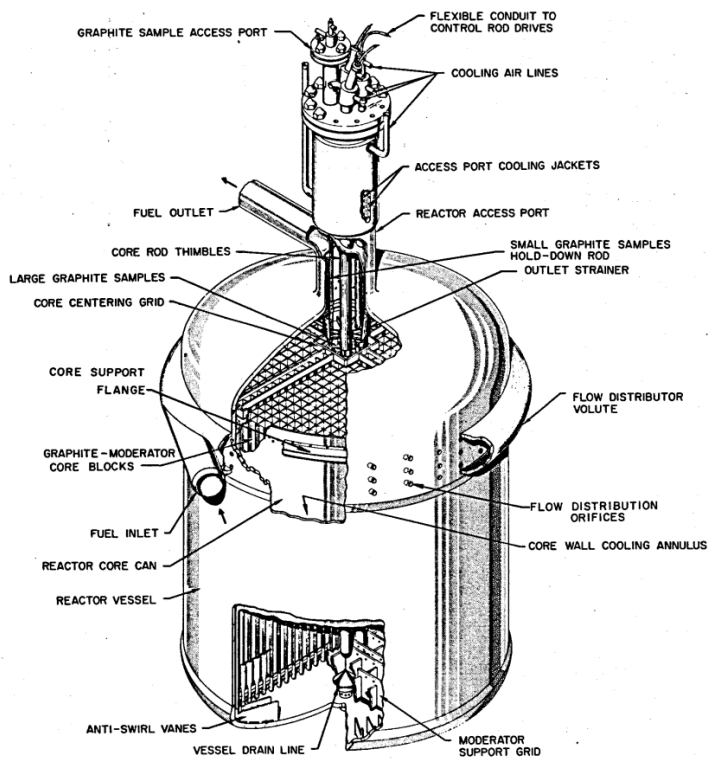


Figure 1. Schematics of the MSRE [2].

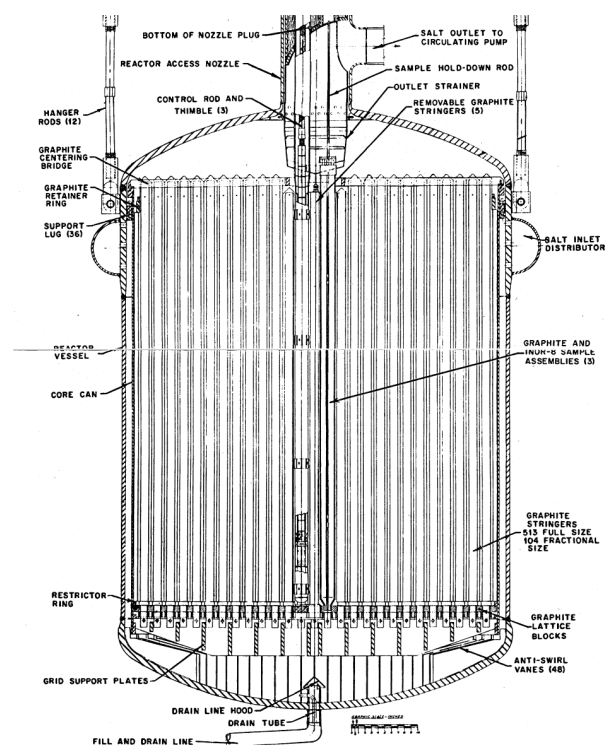


Figure 2. Cross section of the MSRE reactor vessel and core [3].

2. METHODOLOGY

The numerical model of the complete MSRE reactor vessel and its internals is based on the CAD geometry available on the open-source openmsr/msre project [7]. The openmsr model is made considering the high temperature (911 K) dimensions of the reactor vessel. However, fluid dynamic studies of the MSRE were performed at the room temperature on a full-scale mock-up experimental facility using water as a surrogate fluid and the geometry was adjusted accordingly [2]. In Section 3, the presented velocity profiles were obtained using the water, while the temperature profiles were taken from simulations performed with the actual molten salt.

2.1. Monte Carlo Neutronics

Several solvers for neutron transport are available in foamForNuclear, including point kinetics, diffusion, adjoint diffusion, SP3, and discrete ordinates methods [5]. In the present study, however, the spatial power distribution was obtained using the Serpent2 Monte Carlo code [4]. Serpent2 has the capability to read the same OpenFOAM-based unstructured meshes as foamForNuclear, along with the temperature and the density data, which enables incorporating temperature feedback on cross sections at any chosen spatial resolution.

In Serpent2, a simulation of coupled neutron-photon transport was performed using the ENDF/B-VII.1 nuclear data library. The reference power of 7.4 MW was prescribed. Fig. 3 shows a vertical cross section of the Serpent2 model and Fig. 4 shows a horizontal cross sections of the entire model at the volute centerline, and a close-up of the center of the core the with control rods and the sample basket.

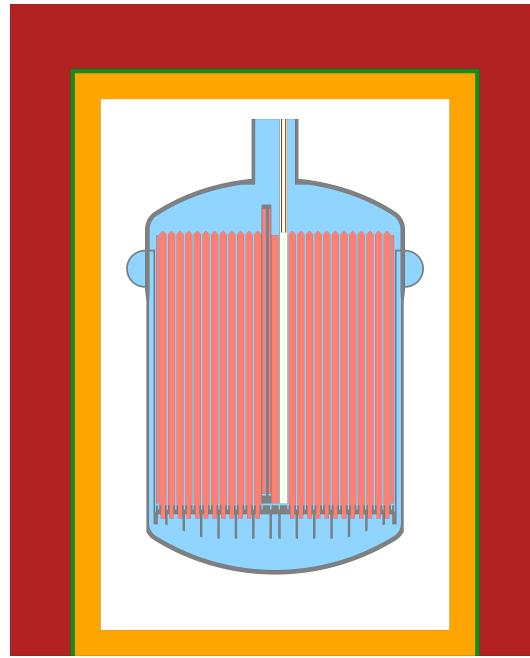


Figure 3. Vertical cross section of the Serpent2 model along the control rods at $y = 5.08$ cm.

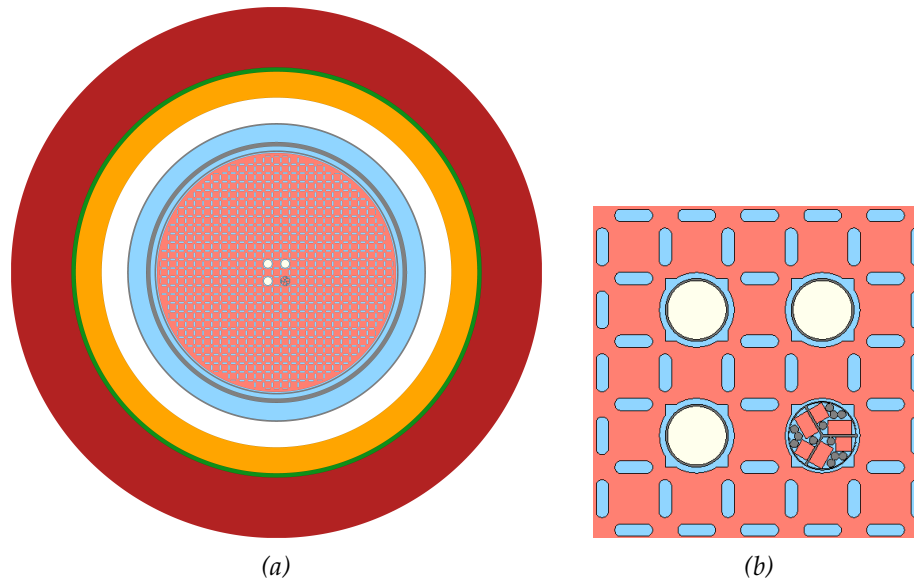


Figure 4. Horizontal cross sections of the Serpent2 model at the volute centerline (a) and the close-up of the core center with the control rods and the sample basket (b).

2.2. Thermal hydraulics

Numerical analyses of conjugate heat transfer and single phase flow, along with volumetric heat generation in the fluid and the solid domain, were performed based on the foamForNuclear multiphysics framework. A compressible Navier–Stokes set of equations with heat transfer and internal heat source terms was used [5], with the $k-\omega$ Shear Stress Transport (SST) model [8] for turbulence closure. Fig. 5 shows the numerical domain used in foamForNuclear.

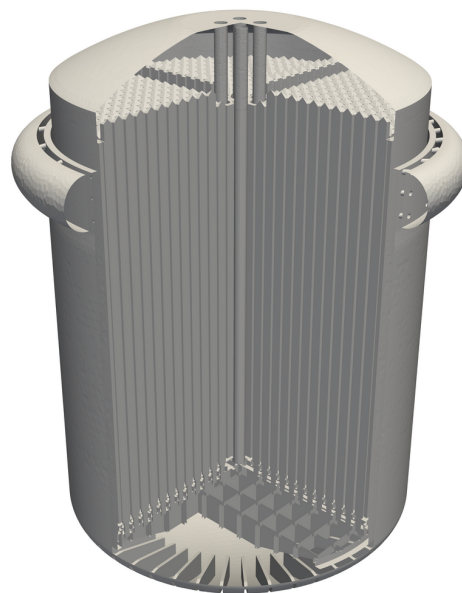


Figure 5. Numerical domain: reactor vessel, lower plenum vanes, grid support plates, graphite lattice blocks, graphite stringers, and control rods.

2.3. Multi-physics coupling

First, a Serpent2 calculation was performed to obtain the spatial power distribution, which was then used in foamForNuclear to obtain the velocity, temperature, and density fields. Next, the temperature and density fields were imported into Serpent2 to obtain new power distributions. This procedure was repeated iteratively until the difference in temperature distributions (obtained with foamForNuclear), affected by differences in power (arising from changes in temperature and density), was reduced to below 1%.

The calculations were performed on a high-performance computing (HPC) system. The first three Serpent2 calculations each required about 12 hours on 56 processors and were used to obtain the spatial power distribution, although with relatively high statistical uncertainty. The final (fourth) calculation was performed with more simulated particles to reduce the uncertainty and required 12 hours on 1024 processors.

The initial foamForNuclear simulation took approximately 12 hours using 768 processors and produced the steady-state solution of the velocity and turbulence fields. Due to a statistically stationary phenomenon, a higher Courant-Friedrichs-Lewy (CFL) of 100 was used. Subsequent runs, initialized from the previously computed fields, required only about 4 hours each to reach the steady state temperature values with CFL number of 500.

3. RESULTS

The fuel enters the reactor vessel through a volute with a constant flow area located approximately at the same height as the top of the reactor core and was then distributed through the orifices into the annular space next to the core wall, as shown in Fig. 6. The orifices were arranged to ensure a uniform flow distribution in the annulus.

Figure 7 shows the angular distribution of fluid velocity along the volute centerline. The velocity distribution in the volute agrees well with the experimental data. Vertical profiles of velocity magnitudes in the annulus are shown in Fig. 8. Because the exact sampling location was not specified, multiple

profiles are plotted every 10° around the circumference. The simulation results indicate a slightly non-uniform flow distribution, in contrast to the experimental observations, which showed a more uniform flow [2].

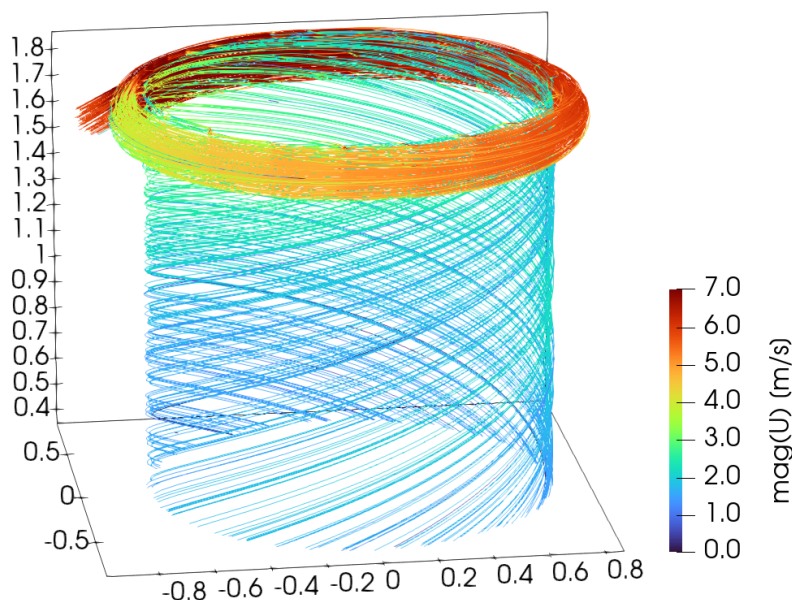


Figure 6. Streamlines in the volute and in the downcomer.

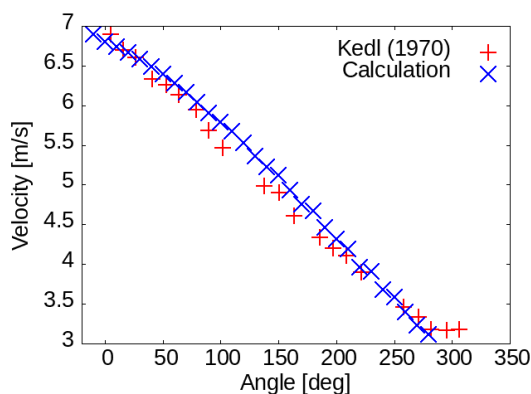


Figure 7. Angular distribution of fluid velocity at the center line of the volute.

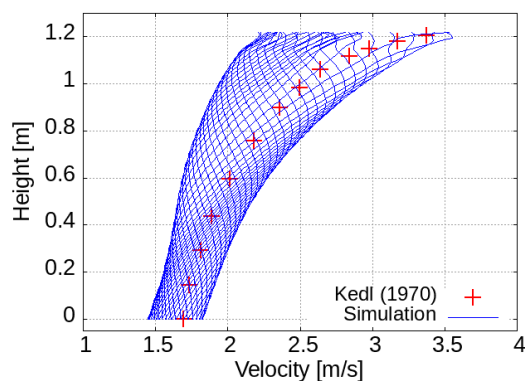


Figure 8. Comparison of vertical profiles of velocity magnitudes in the annulus.

The swirl generated in the volute and the orifices persists through the entire annulus, until it is straightened by the radially placed vanes in the vessel lower plenum, as shown in Fig. 9. Interestingly, the slightly non-uniform flow generated in the volute and in the annulus, generates an off-axis recirculation zone.

Figure 10 shows the volumetric flow rate in the reactor core fuel passages. The arrangement of the plates, which hold the graphite stringers in place, creates larger flow areas below the north-south oriented fuel channels affecting the flow through the core. The simulation results for the north-south (NS) channels agree with the experimental data within one standard deviation, while the discrepancies in the results for the west-east (WE) channels are slightly larger. The average volumetric flow rate obtained from the simulations was $6.77 \times 10^{-5} \text{ m}^3/\text{s}$ for the NS channels and $6.31 \times 10^{-5} \text{ m}^3/\text{s}$ for the WE chan-

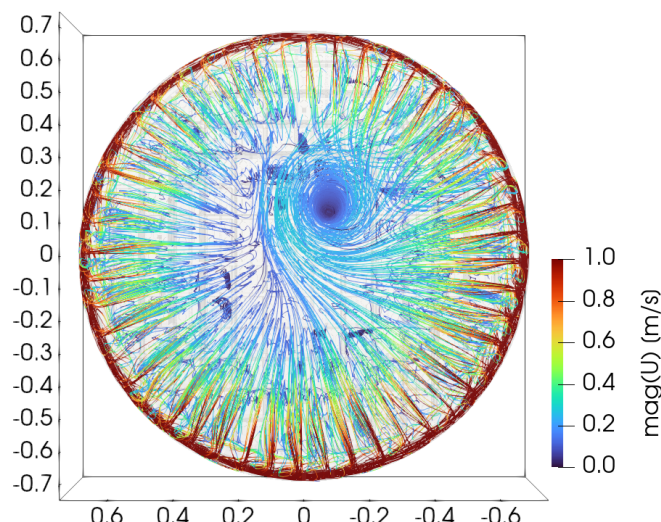


Figure 9. Streamlines in the lower plenum.

nels. In comparison, the experimental measurements reported average flow rates of $6.98 \times 10^{-5} \text{ m}^3/\text{s}$ and $5.47 \times 10^{-5} \text{ m}^3/\text{s}$ for the NS and WE channels, respectively [2]. Unlike the experimental observations, which indicated a pronounced difference in flow rate between channels of different orientations (21%), the simulations did not exhibit such strong directional dependence (7.3%).

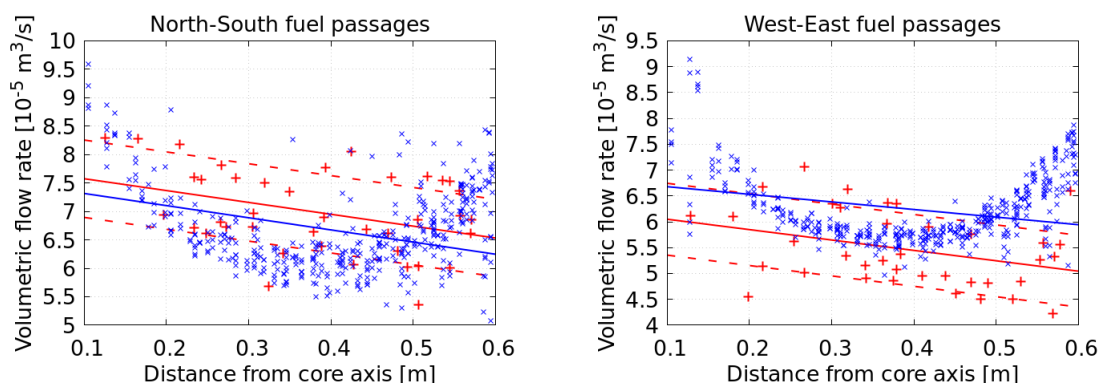


Figure 10. Radial volumetric flow rate distribution in the reactor core fuel passages. Comparison of experimental results (in red) vs. the calculation (in blue). Solid lines represent the least square fit of the data, and dashed lines mark the standard deviation of the experimental results.

Figure 11 shows the spatial power distributions in the fuel and the graphite. The maximum fuel power occurs slightly off-center, due to the presence of the control rod thimbles and the sample basket at the core center, which locally reduce the fuel volume. On the other hand, the peak power within the graphite is reached at the core center.

Figure 12a shows the axial temperature profiles of the fuel and the graphite in the hottest channel. The experimentally measured temperature data for the MSRE are not available. Reported temperature distributions were derived from nuclear analyses employing a simplified twenty-region representation of the reactor core [9]. The graphite temperatures obtained with our model are slightly lower (5 K), but the fuel temperatures differ within a few Kelvin. Fig. 12b shows the spatial temperature distribution. The highest temperature values are reached below the top of the core, at approx. $z = 1.7 \text{ m}$, and off center at $r = 0.3 \text{ m}$. The presence of the control rods and the sample basket decreases the amount of

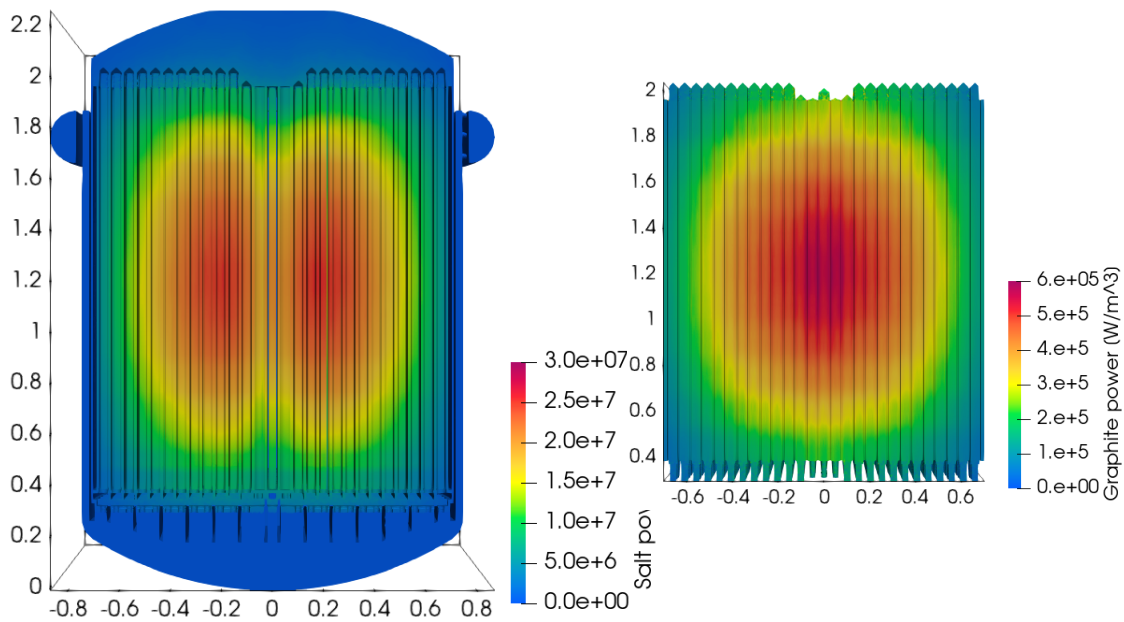


Figure 11. Power distribution in the fuel (figure left) and in the graphite (figure right).

fuel in the core center, while larger fuel passages around these components increase the mass flow rate, and thus reducing the temperature.

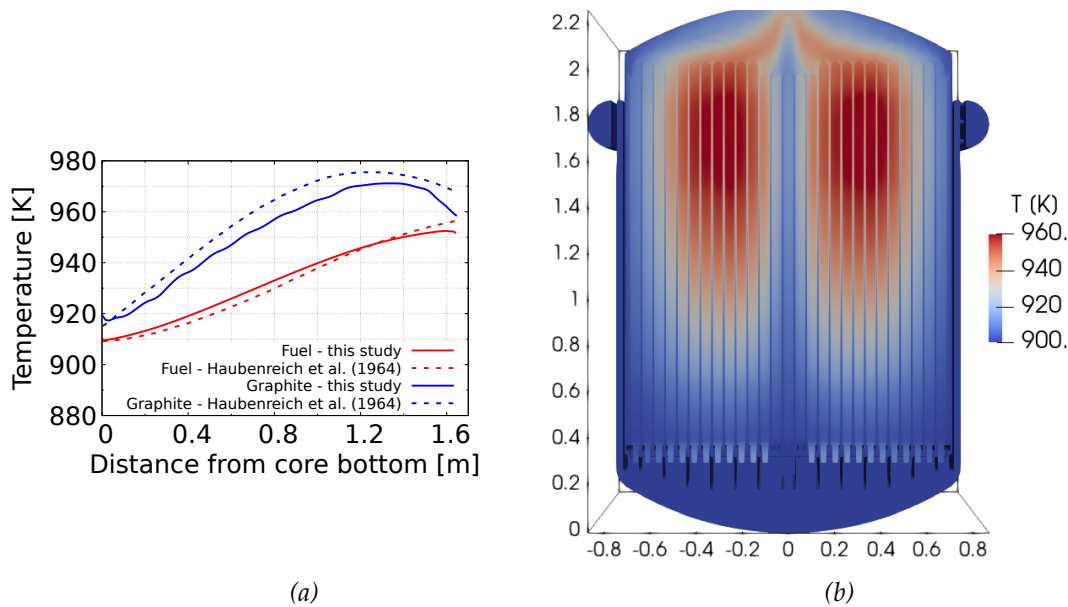


Figure 12. Axial temperature profiles in the fuel and the graphite (a) and temperature distribution (b).

4. CONCLUSIONS

A hyper-fidelity heterogeneous model of the Molten Salt Reactor Experiment was generated, considering the entire reactor vessel. A Monte Carlo simulation using Serpent2 code was performed to obtain the spatial power distributions, and foamForNuclear multiphysics framework code was used to per-

form conjugate heat transfer calculations to obtain the velocity, temperature and density fields, which were then used again in Serpent2 calculation. This iterative procedure was repeated four times, until the difference in the power distributions affected by the temperature and density was sufficiently low. All calculations were performed on a high-performance computing (HPC) system, with each iteration completed within an overnight runtime.

The spatial power distribution, and temperature and velocity fields agree well with the available experimental data. Despite a slightly non-uniform flow in the annulus (downcomer), when compared to the experiment, the discrepancies do not seem to affect the flow in the reactor core.

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